

ISD TERM PROJECT

Software Requirements Specifications

UniVerseX



Project Advisor

Ayesha Zaheer

Presented by:

L1F24BSCS0494	Sania Nasir
L1F24BSCS0340	Seerat Fatima
L1F24BSCS0506	Sharjeel Ahmad

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INTRODUCTION AND BACKGROUND

UniVerseX is a mobile application designed to create a social platform for university students to communicate, share knowledge, and conduct peer-to-peer transactions. Currently, students' discussions scatter across WhatsApp, Facebook, and Instagram, making important information difficult to retrieve and maintain continuity. Our platform consolidates academic discussions, social interaction, and student commerce into a single, structured environment accessible on both iOS and Android devices.

1.1. Problem Statement

University students face fragmented communication channels where academic discussions, social posts, and marketplace activity are scattered across multiple unstructured platforms. This fragmentation results in:

- Loss of important academic information in group chats
- Difficulty finding answers to repeated questions
- Unsafe, unmoderated student commerce
- No dedicated space for campus-specific discussions

UniVerseX solves this by providing a unified platform where students can:

- Post and discuss course-related topics in moderated communities
- Share confessions and social content anonymously
- Buy, sell, and barter textbooks, notes, and campus accessories
- Access all information within a single, organized interface in various dedicated communities

1.2. Background

In universities, communication is essential for academic success and campus life. Students currently rely on general-purpose social media platforms not specifically designed for academic purposes. WhatsApp groups become cluttered, Facebook groups are public and unprofessional, and Instagram stories fade within 24 hours. There is a clear demand for a dedicated student platform that is community-driven.

The proposed solution draws inspiration from successful platforms: **Reddit**, **Piazza** and **Empor** remaining lightweight and prototype-focused using **MIT App Inventor**.

The platform will emphasize moderation, user safety, and organized architecture to ensure a healthy discussion environment.

1.3. Scope

UniVerseX will be developed as a cross-platform mobile application (iOS/Android) with the following core features:

Authentication & Profile:

- Email/password registration and login
- User profile with username and reputation score
- Profile view showing user's posts and marketplace listings

Discussion Features:

- Create text-based posts with optional image uploads
- Post categorization (Question, Discussion, Confession)
- Moderated communities
- Comment threads under posts
- Upvote/downvote mechanism for post visibility
- Search posts by keyword, category, or community (Limited)

Marketplace Features:

- List items for sale or barter (books, notes, accessories, etc.)
- Contact seller through in-app messaging
- User ratings for marketplace transactions
- Item details with images, price, and barter options

Moderation:

- Community moderators (admins) appointed per community
- Ability to report inappropriate content
- Admin dashboard for content removal
- Spam and abuse prevention

Technical Stack:

- Frontend: MIT App Inventor (cross-platform blocks)
- Backend: CloudDb (Realtime Database + Authentication)
- Image Storage: Cloudinary (free tier)
- Target: Android + iOS

1.4. Objectives

- Provide a centralized, organized platform for student communication and knowledge sharing
- Demonstrate practical application of software engineering principles in a real-world project
- Eliminate information fragmentation across multiple apps
- Create safe, moderated spaces for academic and social discussions
- Improve student engagement and sense of campus community
- Enable peer-to-peer student commerce with built-in trust mechanisms

1.5. Challenges

Real-time Synchronization: Ensuring feed updates reflect new posts and comments instantly across users

Database Complexity: Managing relationships between users, posts, comments, and marketplace listings

Image Handling: Optimizing image uploads and storage within app size constraints

User Trust: Building confidence in marketplace transactions (ratings, reviews, dispute resolution)

Performance: Managing and organizing a large number of posts and comments

MIT App Inventor Limitations: Lack of advanced features like *nested* queries and complex business logic

1.6. Learning Outcomes

Upon completion, team members will have gained:

- Understanding of *Software Requirements Specification* documentation
- Mobile app development using MIT App Inventor and visual programming
- Cloud database design and real-time synchronization with Firebase
- Team collaboration and version control (document management)
- ***Software engineering lifecycle*** from specification to prototype

1.7. Nature of End Product

UniVerseX will be delivered as a **functional prototype** demonstrating core functionality:

- Cross-Platform app
- Fully functional discussion and post features
- Working marketplace with image uploads
- Real-time Firebase backend integration
- Moderated community system
- User authentication and profile management

The prototype prioritizes functionality over polish; advanced features like performance optimization and complex analytics are **excluded**.

1.8. Completeness Criteria

The project will be considered complete when:

1. Users can register, log in, and maintain profiles
2. Posts can be created, displayed, and deleted by authors
3. Comments appear under posts in real-time
4. Upvote/downvote system functions correctly and reflects in post scoring
5. Communities can be created and moderated
6. Marketplace listings can be posted with images and descriptions
7. Search functionality returns relevant results
8. Feed displays posts sorted by newest and popularity
9. Marketplace transactions show contact options

1.9. Business Goals

- Increase student engagement within the university ecosystem
- Provide a safe alternative to unmoderated third-party platforms
- Reduce dependency on external social media for campus discourse
- Enable sustainable student commerce with built-in buyer protection
- Build a foundation for potential university adoption and expansion

1.10. Related Works

Reddit (reddit.com): Community-driven discussion platform with upvote/downvote ranking, moderated subreddits, and user reputation. UniVerseX adopts the moderated community model and post-based discussion format.

Piazza (piazza.com): Academic Q&A platform used in universities with instructor monitoring and structured answers. UniVerseX incorporates the educational focus and moderation hierarchy.

OLX / Facebook Marketplace: Peer-to-peer marketplace platforms. UniVerseX integrates marketplace functionality with trust ratings and in-app messaging.

UniVerseX differentiates itself by combining all three elements (*discussion, academic focus, and marketplace*) in one platform specifically designed for universities, rather than adapting general-purpose tools.

1.11. Document Conventions

Key Terms: User, Post, Comment, Community, Moderator, Listing, Marketplace are used consistently throughout

Terminology: "Post" refers to discussion content; "Listing" refers to marketplace items

Hierarchy: Communities contain Posts; Posts contain Comments.

OVERALL DESCRIPTION

2.1. Product Features

The major features of the system are listed below:

A. Discussion & Posts

- Create posts with title, body text, tags and optional images
- **Tags:** Question, Discussion, Confession, etc. to improve organization and searchability
- Upvote/downvote mechanism with score calculation
- Comment threads on each post
- Edit and delete own posts
- Pin important posts (moderators only)
- Search posts by keyword, tag, or community

B. Communities

- Create communities around topics, courses, or interests
- Join communities
- View community-specific feed
- Moderator appointment and role management
- Community moderation tools (pin, remove posts)
- Community rules and description

C. Marketplace

- List items for sale or barter with images
- Item categories (Books, Notes, Accessories, Electronics, etc.)
- Seller contact information and ratings
- In-app messaging between buyer and seller
- Transaction status tracking (Available, Sold, Exchanged)
- User reputation/ratings system

D. User & Authentication

- Email/password registration and login (**must be university mail**)
- Profile customization (username, bio, profile picture)
- View user's posts and marketplace listings
- Reputation score (based on upvotes and transaction ratings)
- Follow users (optional, for future phases)

E. Moderation & Safety

- Report inappropriate content
- Content removal and user warnings
- Spam detection and prevention
- Transaction dispute resolution tools
- Moderator dashboard per community

2.2. User Classes and Characteristics

1. Regular Students (Primary Users)

- University students aged 18 – 25
- Basic to intermediate mobile app experience
- Use the app for academic discussions, social posts, and buying/selling items
- Expect fast, intuitive interface with minimal learning curve
- May create 5 – 10 posts per week on average

2. Moderators (Secondary)

- Experienced users appointed by admins or community creators
 - Responsibility:* monitor content, remove spam, enforce community guidelines
- Higher privileges: ability to pin/remove posts, warn users
 - Typically 1 – 2 per community of 100+ members

3. Administrators (System Admins)

- Full platform access: view all posts, remove any content, manage moderators
- *Italics* indicate special role; appointed by development team
- Ability to disable accounts and enforce platform-wide policies

4. Guest Users (Optional, Future Phase)

- Non-registered users who can view public posts (read-only)
- Cannot create posts, comment, or access marketplace
- Encouraged to register for full experience

2.3. Operating Environment

Platform Specifications:

- **Mobile OS:** Android 8.0+ and iOS 8.0+
- **Network:** Requires active internet connection (WiFi or mobile data)
- **Hardware:** Minimum 2GB RAM, 50MB storage

Development & Deployment:

- **Development Tool:** MIT App Inventor (visual blocks programming)
- **Backend:** CloudDb (Realtime Database + Authentication + Hosting)
- **Testing:** Android emulator + physical device testing
- **Image Storage:** Cloudinary

Browser Requirements:

- Modern browsers supporting Service Workers
- **HTTPS** required for all communications

2.4. Design and Implementation Constraints

MIT App Inventor Limitations:

- Limited to visual block programming (no direct code)
- No native support for complex nested queries
- Maximum app size ~50MB after compilation

CloudDb Free Tier:

- 100 concurrent connections maximum
- 1GB total storage
- Rate limiting on reads/writes

Image Handling:

- Cloudinary free tier limited to 10GB/month bandwidth
- Images must be optimized before upload

Cross-Platform Support:

- Native Android recommended for full feature parity
- Testing required on both platforms despite code reuse

Development Scope:

- Scalability features (caching, CDN, load balancing) excluded
- Advanced analytics and machine learning not included
- Real-time notifications may have delays
- Project prototype status; *not production-ready*

2.5 Assumptions and Dependencies

Assumptions:

- All users have access to mobile devices and have internet connectivity
- Users possess basic mobile app literacy
- University emails are available for authentication
- CloudDb services remain available throughout project
- Cloudinary API availability for image uploads

Dependencies:

- **MIT App Inventor:** Platform for development and compilation
- **Google Account:** Required to access Firebase and MIT App Inventor
- **Internet Services:** Continuous connectivity for real-time sync

External Risks:

- Dependency on CloudDb availability and pricing changes
- MIT App Inventor updates may affect compilation
- Cloudinary service disruptions

Project Management

3.1. Work Breakdown Structure

➤ **1.0 Initiation & Requirements**

- 1.1 Problem Statement Analysis
- 1.2 System Requirements Specifications (SRS)

➤ **2.0 Design & Architecture**

- 2.1 UI/UX Screen Wireframing
- 2.2 CloudDb Database Schema Mapping

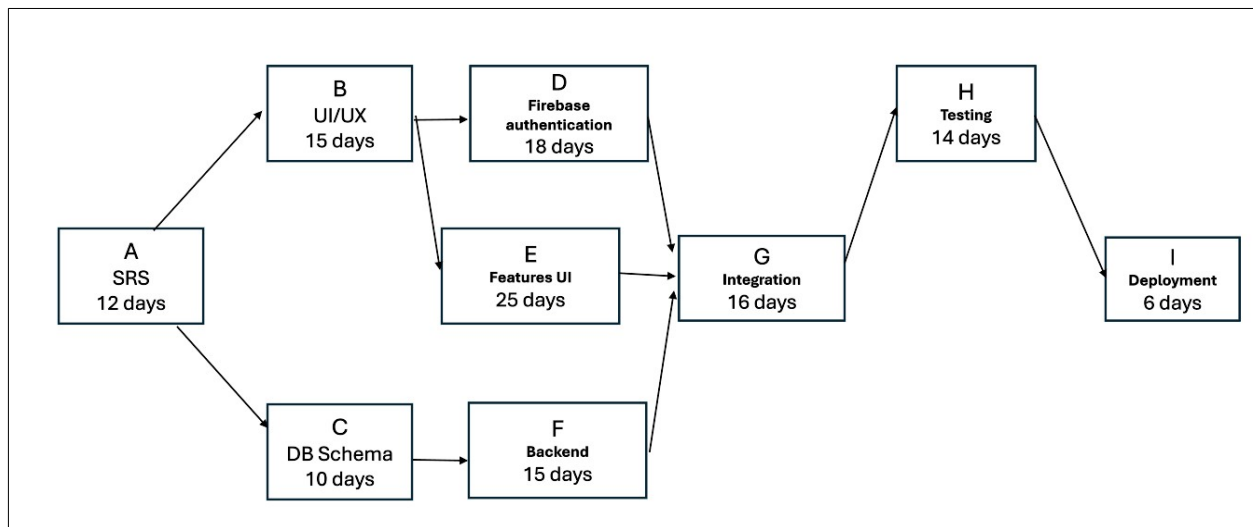
➤ **3.0 Core Application Development**

- 3.1 Authentication and Profile Blocks
- 3.2 Discussion Feed & Comment Threads
- 3.3 Marketplace Listings Layout
- 3.4 Community Moderation Controls

➤ **4.0 Verification & Deployment**

- 4.1 Integration Testing via Android Emulator
- 4.2 Final Prototype Sign-off

3.2. Activity Graph

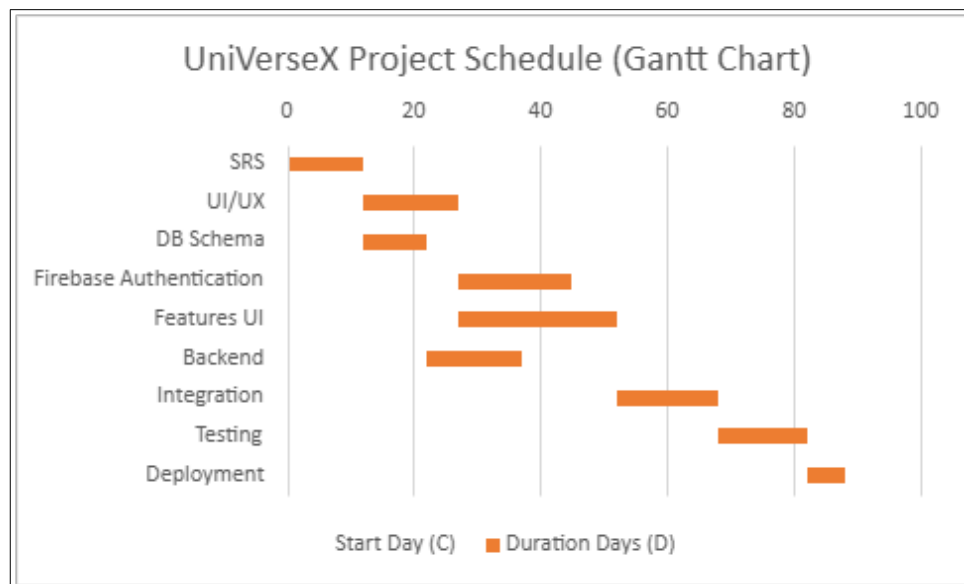


3.3. Critical Path

Path	Sequence of Activities	Total Duration
Path 1	A → B → E → G → H → I	88 days
Path 2	A → B → D → G → H → I	81 days
Path 3	A → C → F → G → H → I	73 days

Critical path: A → B → E → G → H → I (88 days)

3.4. Gantt chart



FUNCTIONAL REQUIREMENTS

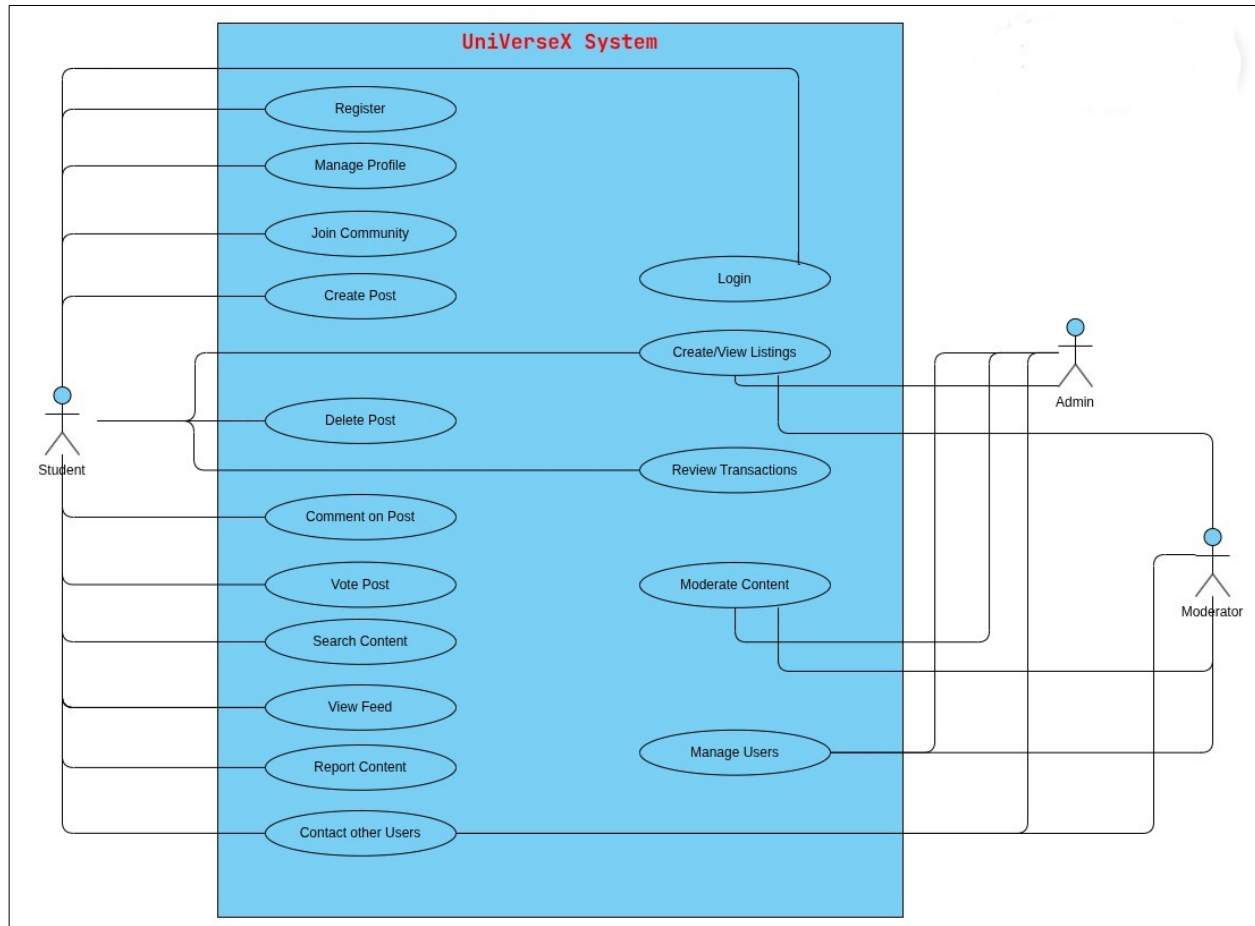
4.1. Requirements Analysis and Modeling

Actors:

- **User (Regular Student):** Primary actor interacting with discussion and marketplace features
- **Admin (System Administrator):** Platform-wide moderation and management
- **Moderator (Community Moderator):** Community-specific content management

Main Use Cases:

1. Register Account
2. Login to Account
3. Manage Profile
4. Create/Join Community
5. Create Post
6. Comment on Post
7. Upvote/Downvote Post
8. Search Posts/Content
9. Create Marketplace Listing
10. Contact Seller / In-App Messaging
11. Rate/Review Transaction
12. Report Content
13. Moderate Content (Remove Posts/Users)
14. View Community Feed
15. Delete Own Post

Use-Case Diagram:

4.2. Use Cases

1. Create/Join Community

Identifier	UC - 01	
Purpose	To allow users to join/create communities	
Actor	Student	
Priority	High	
Pre-conditions	User must be logged in	
Post-conditions	➤ User successfully joins an existing community OR ➤ User creates and becomes admin of a new community	
Typical Course of Action		
S#	Actor Action	System Response
1	User views communities	System shows list of available communities
2	User selects community OR chooses “Create Community”	System shows details OR creation form
3	User clicks “Join”	System adds user as a member
4	User enters community details (for creation) and submits	System creates new community and assigns user as admin
Alternate Course of Action		
S#	Actor Action	System Response
1	User tries to join a private community	System requests approval or sends join request
2	User is already a member	System displays “ Already a member ” message
3	User submits incomplete/invalid creation form	System shows error and asks for corrections
4	Community creation fails (e.g., duplicate name)	System displays failure message and prompts retry

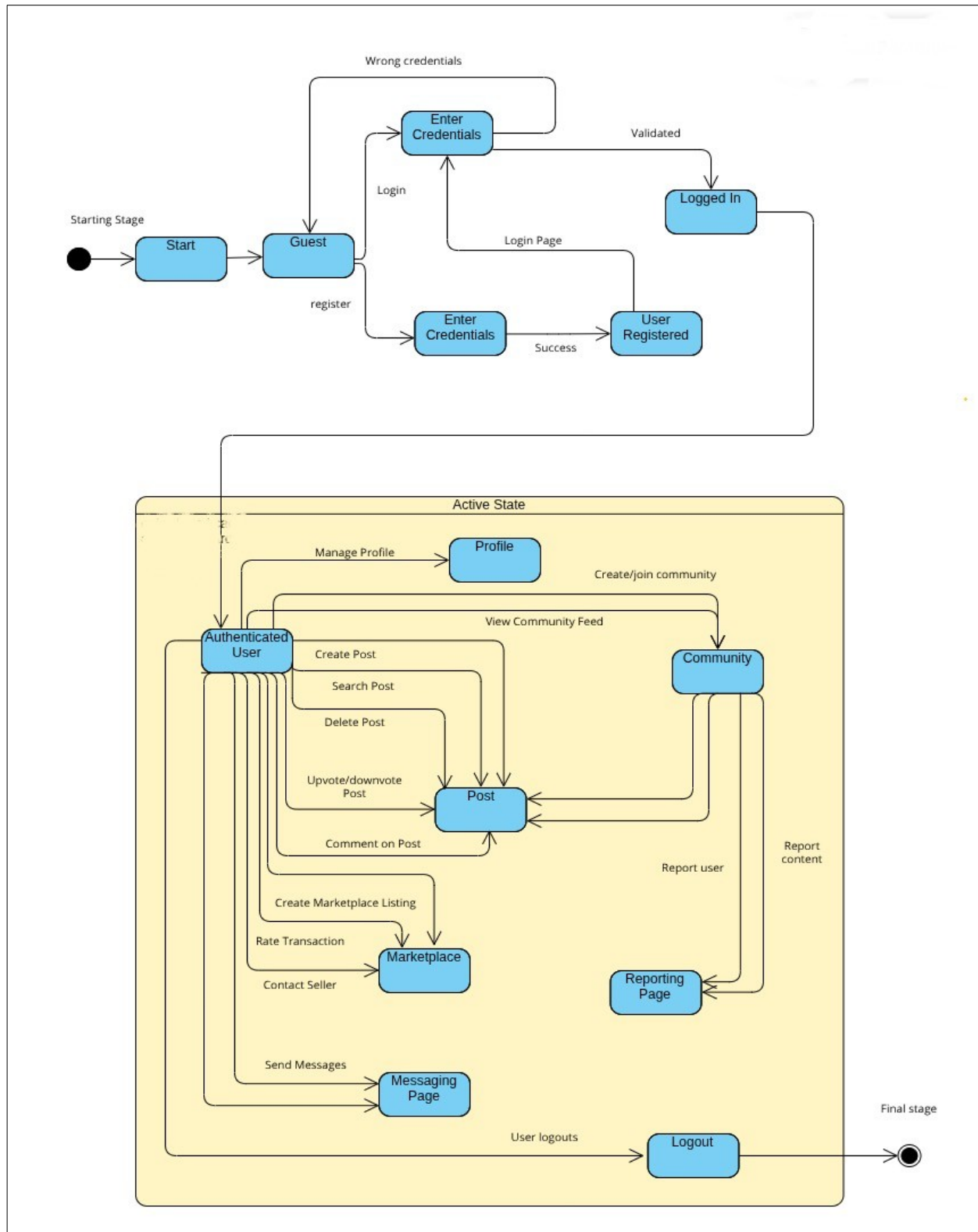
2. Create Post

Identifier	UC - 05	
Purpose	To allow users to publish posts on profile or in communities.	
Actor	Student	
Priority	High	
Pre-conditions	User must be logged in.	
Post-conditions	Post is published.	
Typical Course of Action		
S#	Actor Action	System Response
1	User selects create post	System opens editor
2	User enters content	System validates
3	User submits post	System publishes
Alternate Course of Action		
S#	Actor Action	System Response
1	Empty post submitted	System rejects and show error
2	Unsupported file format uploaded	System denies upload
3	Network failure during submission	System does not publish post and show retry message

3. Create Marketplace Listing

Identifier	UC - 03	
Purpose	To allow users to sell or barter items	
Actor	Student	
Priority	Medium	
Pre-conditions	User logged in	
Post-conditions	Listing or transaction is processed	
Typical Course of Action		
S#	Actor Action	System Response
1	User opens marketplace	System shows listings
2	User clicks create listing	System shows form
3	User submits details	System publishes listing
Alternate Course of Action		
S#	Actor Action	System Response
1	Invalid listing details	System rejects listing submission

4.3. State Diagram



State Diagram Info:

The state diagram of UniVerseX represents the different states a user or system component transitions through during interaction with the application. It illustrates how the system responds to user actions such as logging in, creating posts, interacting with content, and logging out. Initially, the system starts in an ***Unauthenticated State***, where the user can either register or log in. Upon successful authentication, the user transitions to the *Authenticated State*, gaining access to features like viewing the feed, joining communities, creating posts, commenting, and interacting with the marketplace. Each action (e.g., creating a post, sending a message, or listing an item) represents a transition that leads to corresponding states such as *Post Created*, *Comment Added*, or *Listing Published*. The system also handles alternate transitions like errors (invalid input, network failure) and moderation actions (content removal), ensuring proper flow control. Finally, the user can transition to the *Logged Out State*, returning the system to its initial state. This diagram helps visualize system behavior, ensuring all possible user interactions and system responses are clearly defined and managed.

Non-Functional Requirements

This section defines the quality attributes of the UniVerseX system. These requirements ensure that the system performs efficiently, remains secure, and provides a smooth user experience for students.

5.1. Performance Requirements

The system is designed to support real-time interaction and must respond quickly to user actions.

- The system shall load the main feed within **2–3 seconds** under normal conditions.
- The system shall reflect new posts and comments in **real-time or near real-time**.
- The system shall support multiple active users without noticeable delay.
- The system shall handle image uploads efficiently without slowing down the application.

5.2. Safety Requirements

The system must provide a safe environment for users, especially due to user-generated content and marketplace interactions.

- The system shall allow users to report inappropriate or abusive content.
- The system shall enable moderators to take action against reported content.
- The system shall prevent misuse of the platform through moderation controls.
- The system shall ensure safe interaction between buyers and sellers.

5.3. Compatibility Requirements

The system should work across supported devices and platforms.

- The system shall run on Android and iOS devices.
- The system shall function on devices with basic hardware requirements.
- The system shall require an active internet connection for operation.

5.4. Security Requirements

Security is essential to protect user data and ensure controlled access.

- The system shall allow access only to authenticated users.
- The system shall use secure methods for user authentication.
- The system shall protect user data from unauthorized access.
- The system shall ensure secure communication between user and system.

5.5. Usability Requirements

The system should be simple and easy to use for university students.

- The system shall provide a clear and user-friendly interface.
- The system shall allow users to perform key actions easily (posting, commenting, searching).
- The system shall provide feedback messages for user actions.
- The system shall maintain consistency across all screens.

5.6. Reliability Requirements

The system must remain stable and dependable during usage.

- The system shall operate with minimal downtime.
- The system shall ensure that user data is not lost during failures.
- The system shall handle errors without crashing.
- The system shall confirm successful completion of user actions.

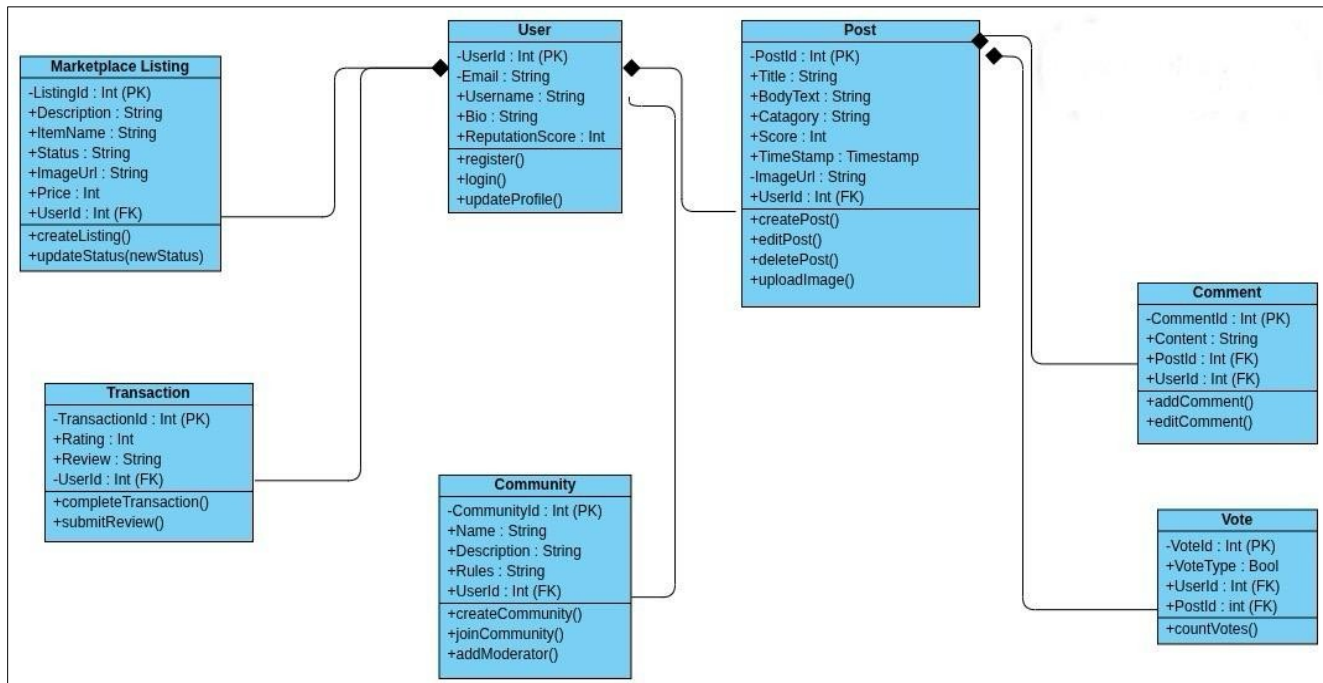
5.7. Scalability Requirements

The system should support future growth and expansion.

- The system shall handle an increasing number of users and posts.
- The system shall support future feature enhancements.
- The system shall allow integration with more advanced services if required.

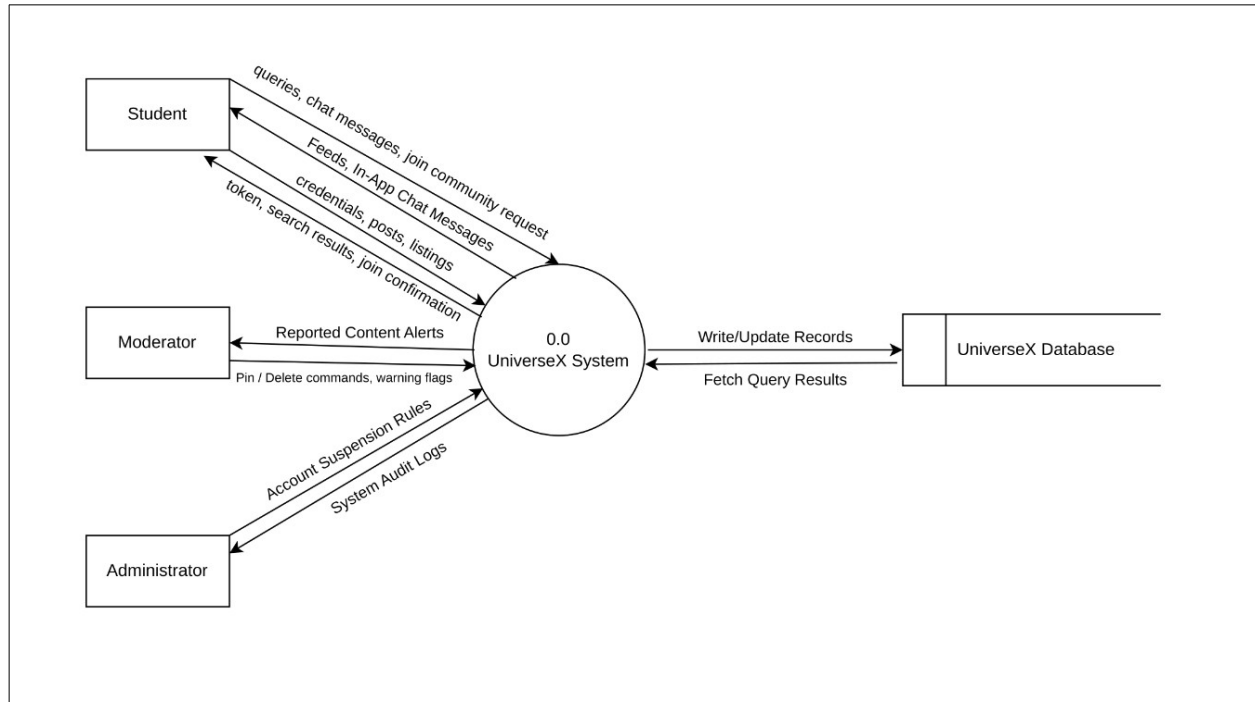
Designing

7.1. Class Diagram

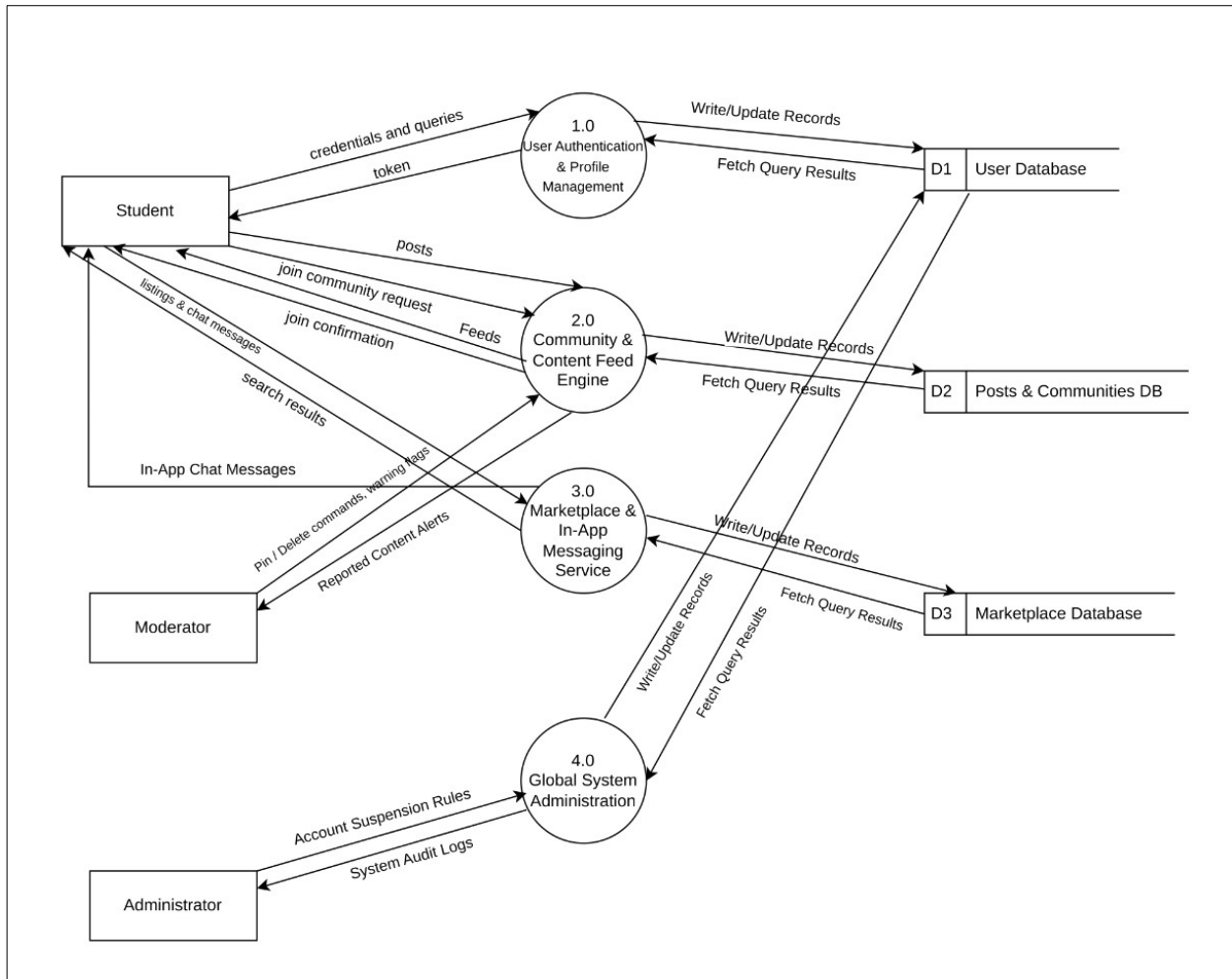


7.2. Data Flow Diagram

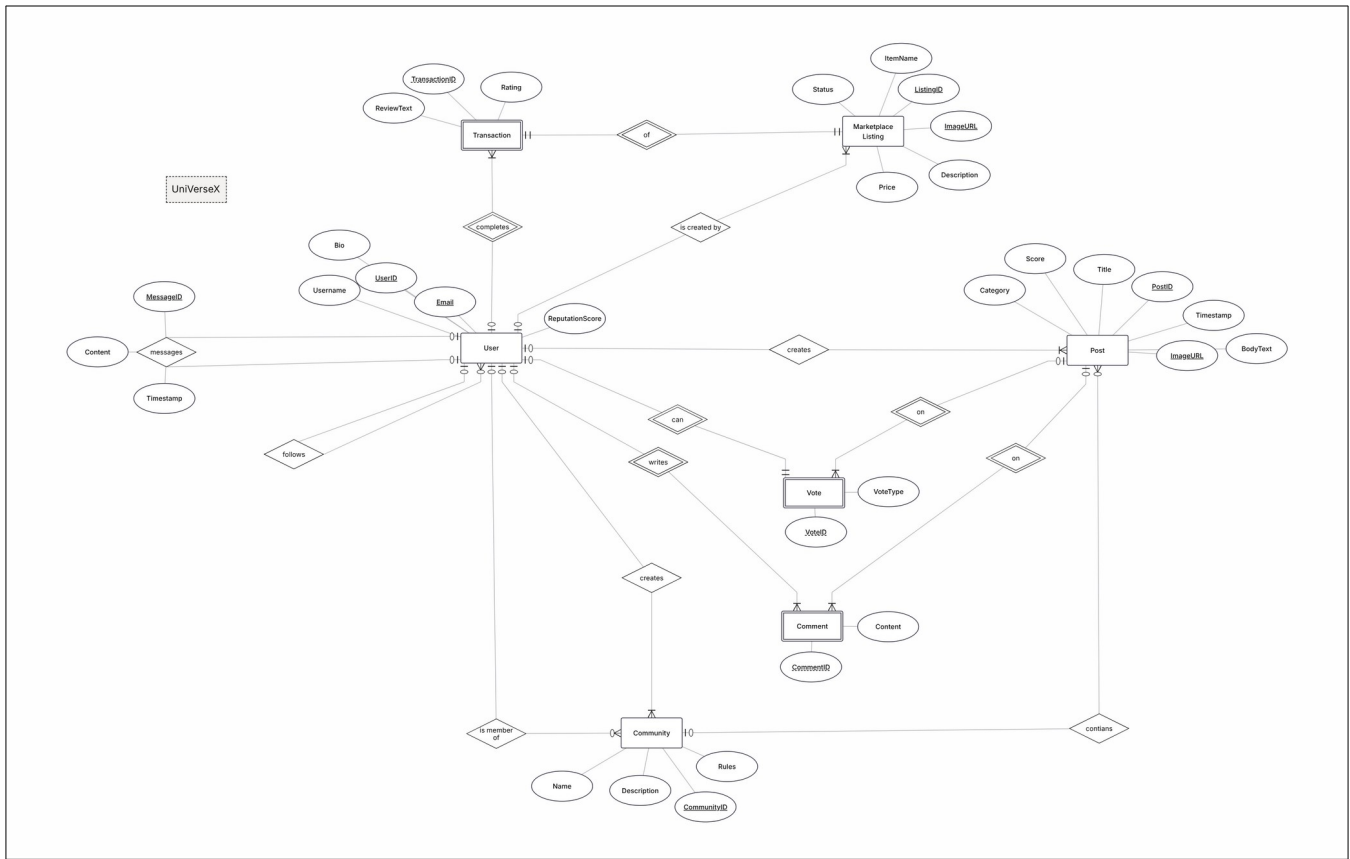
7.2.1. DFD – (Level – 0)



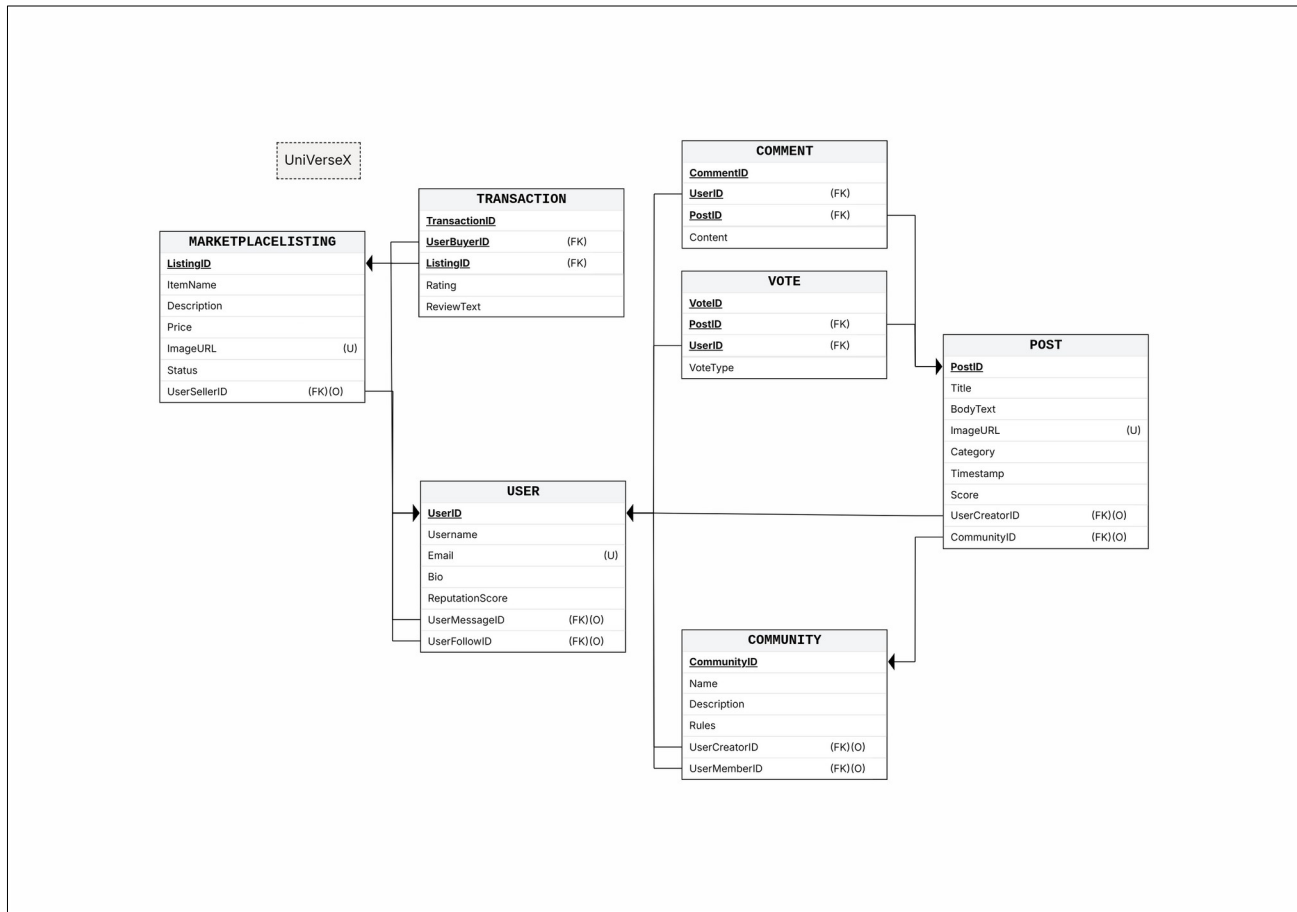
7.2.2. DFD – (Level – 1)



7.3. ER Diagram

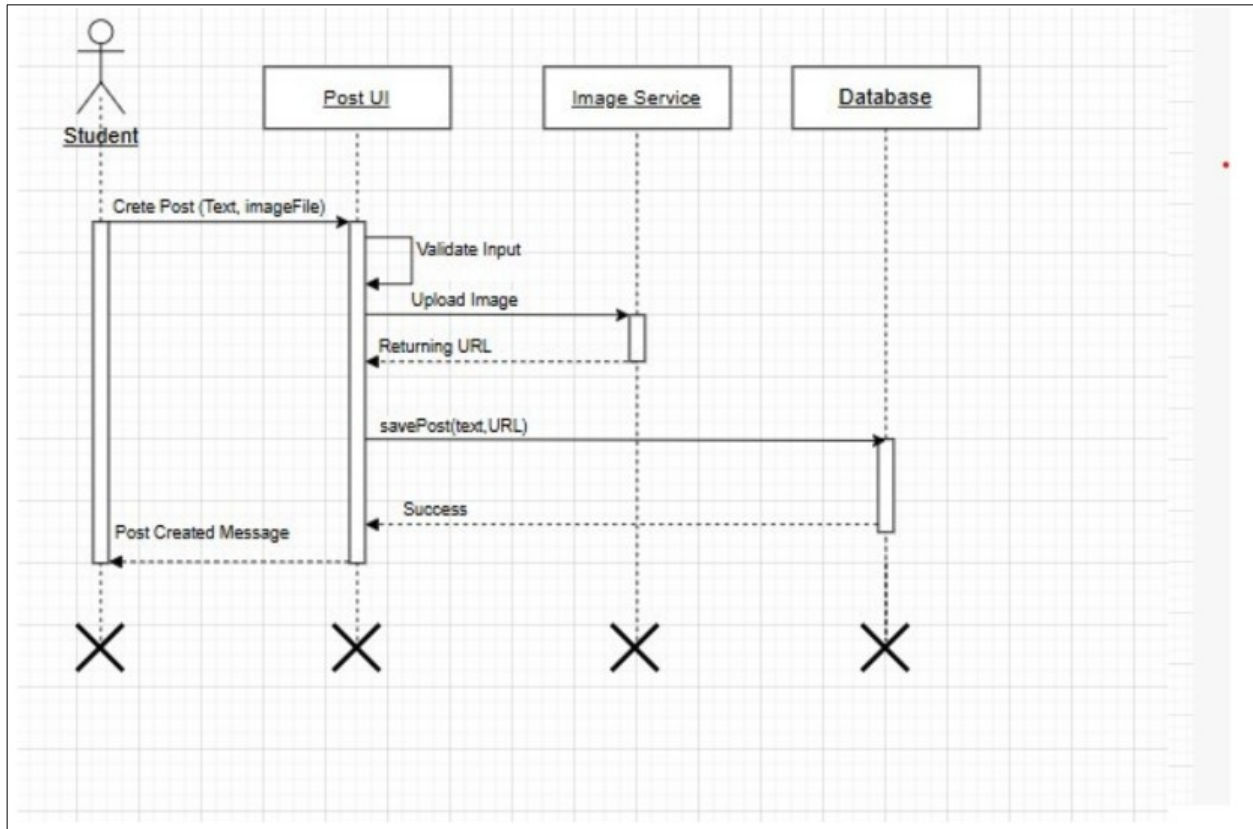


7.4. Physical design of database

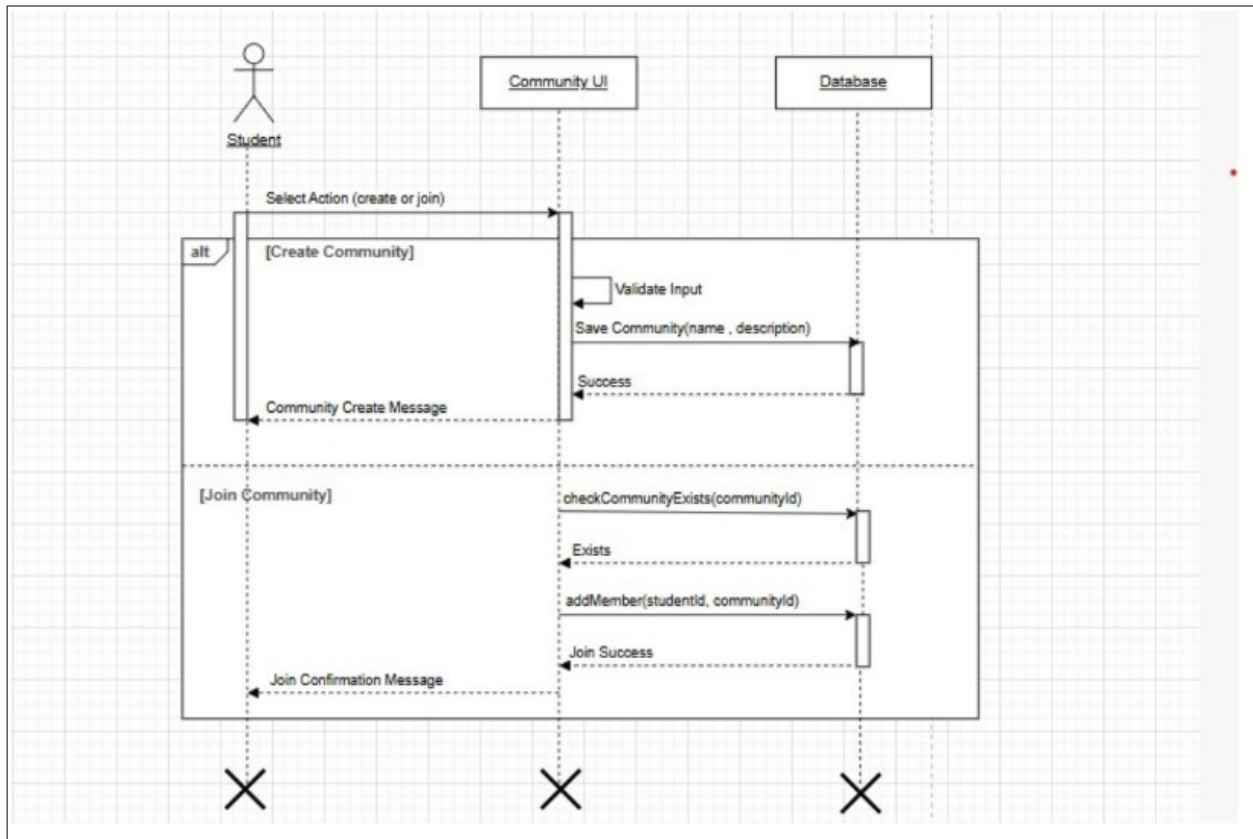


7.5. Sequence Diagrams

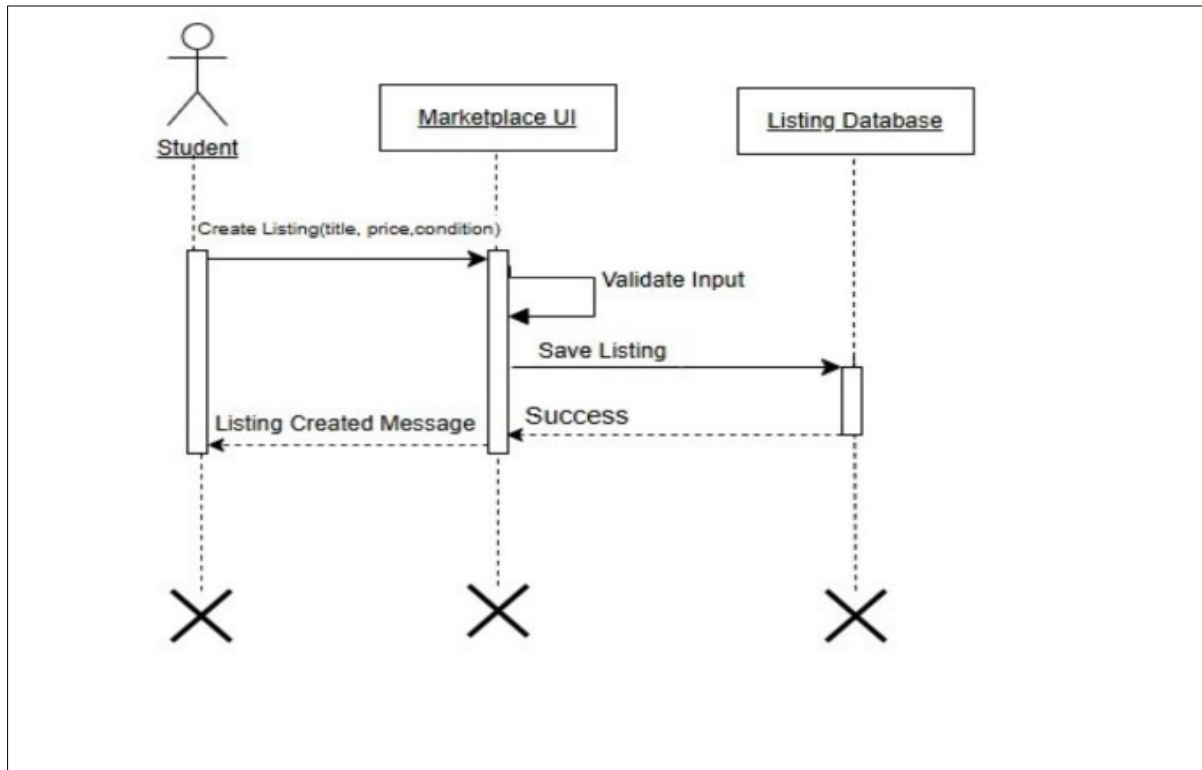
- 7.5.1. Create Post



- 7.5.2. Create Community



- 7.5.3. Create Marketplace Listing



Estimation

8.1. Cost Benefit Estimation

Estimated Development Cost (1st Year)

Item	Estimated Cost (PKR)
Development Effort	150,000
Testing & Maintenance	30,000
Cloud Services & Storage	20,000
Miscellaneous Expenses	10,000
Total Cost	210,000

Estimated Benefits (1st Year)

Benefit Source	Estimated Value
Student Marketplace Transactions	180,000
University Adoption Value	100,000
Advertisement Opportunities	80,000
Increased Student Engagement Value	40,000
Total Benefits	400,000

8.1.1. Return of Investment {1st year}

Total cost= Rs **210,000**

Total benefit= Rs **400,000**

$$ROI = \frac{Benefit}{Cost}$$

$$= \frac{400,000}{210,000}$$

$$= 1.90$$

Interpretation: For every Rs. 1 invested, the project is expected to return approximately Rs. 1.90 during the first year.

8.1.2. %gain on ROI {1st year}

$$\text{Formula of \%gain on ROI} = \left\{ \frac{(Benefits - Cost)}{Cost} \right\} \times 100$$

$$\left\{ \frac{(400,000 - 210,000)}{210,000} \right\} \times 100$$

$$= 90.48\%$$

8.1.3. Payback Period in years

$$\text{Formula of payback period} = \frac{Cost}{Benefit}$$

$$= \frac{210,000}{400,000}$$

$$= 0.53 \text{ Years} \approx 6.4 \text{ Months}$$

Interpretation: The initial investment is expected to be recovered within approximately 6 months.

8.2. FP Based Estimation

Function Point Analysis

Function Type	Count	Weight	FP
External Inputs (Login, Registration, Create Post, Create Listing, Join Community)	5	4	20
External Outputs (Feeds, Profiles, Listings, Search Results)	4	5	20
External Inquiries	3	4	12
Internal Logical Files (Users, Posts, Communities, Listings, Messages)	5	10	50
External Interface Files	2	7	14

Unadjusted Function Points (UFP) = 116

Assuming Value Adjustment Factor (VAF) = 1.05

Adjusted FP = $116 \times 1.05 = 121.8 \approx 122$ FP

8.3. COCOMO Estimation

Step 1: Convert FP to KLOC

For a mobile application:

1 FP $\approx$ 50 LOC

LOC = 122×50

LOC = 6,100

KLOC = 6.1

Step 2: Effort Estimation

$$E = a^b (KLOC)^{b^b}$$

$$E = (2.4)(6.1)^{1.05}$$

$$E \approx 16$$

Effort $\approx$ 16 Person-Months

Step 3: Development Time

$$TDEV = c_b (E)^{d_b}$$

$$= (2.5) \times (16)^{0.38}$$

$$\approx 7$$

Time $\approx$ 7 Months

Step 4: Team Size

$$\text{Team Size} = \frac{\text{Effort}}{\text{Time}}$$

$$= \frac{16}{7}$$

$\approx$ 2–3 Developers

References

- [1] IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications
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